Algebraic substitution

Substituting an expression

Replace a variable with a linear expression, then expand and simplify.

10 MINUTES / 4 QUESTIONS / AS-06

Name _____ Date _____

Show your working. Use brackets for negative values.

Starter: use the full set. Exit ticket: choose 2-3 questions.

1 Given $x = 3y - 2$

Write in terms of y and simplify

Simplify $6x - 7$

2 Given $x = 6y + 5$

Write in terms of y and simplify

Simplify $3x + 3$

3 Given $x = 2y + 1$

Write in terms of y and simplify

Simplify $5x - 2$

4 Given $x = 4y + 4$

Write in terms of y and simplify

Simplify $4x - 7$

Answer sheet

Substituting an expression | 4 questions

Keep this section separate from the question sheet. Full working follows on the next pages.

01 $18y - 19$

02 $18y + 18$

03 $10y + 3$

04 $16y + 9$

Worked solutions

Substituting an expression

01 Given $x = 3y - 2$

$6x - 7$

Substitute $x = 3y - 2$.

$6(3y - 2) - 7$

$18y - 12 - 7$

$18y - 19$

02 Given $x = 6y + 5$

$3x + 3$

Substitute $x = 6y + 5$.

$3(6y + 5) + 3$

$18y + 15 + 3$

$18y + 18$

03 Given $x = 2y + 1$

$5x - 2$

Substitute $x = 2y + 1$.

$5(2y + 1) - 2$

$10y + 5 - 2$

$10y + 3$

04 Given $x = 4y + 4$

$4x - 7$

Substitute $x = 4y + 4$.

$4(4y + 4) - 7$

$16y + 16 - 7$

$16y + 9$